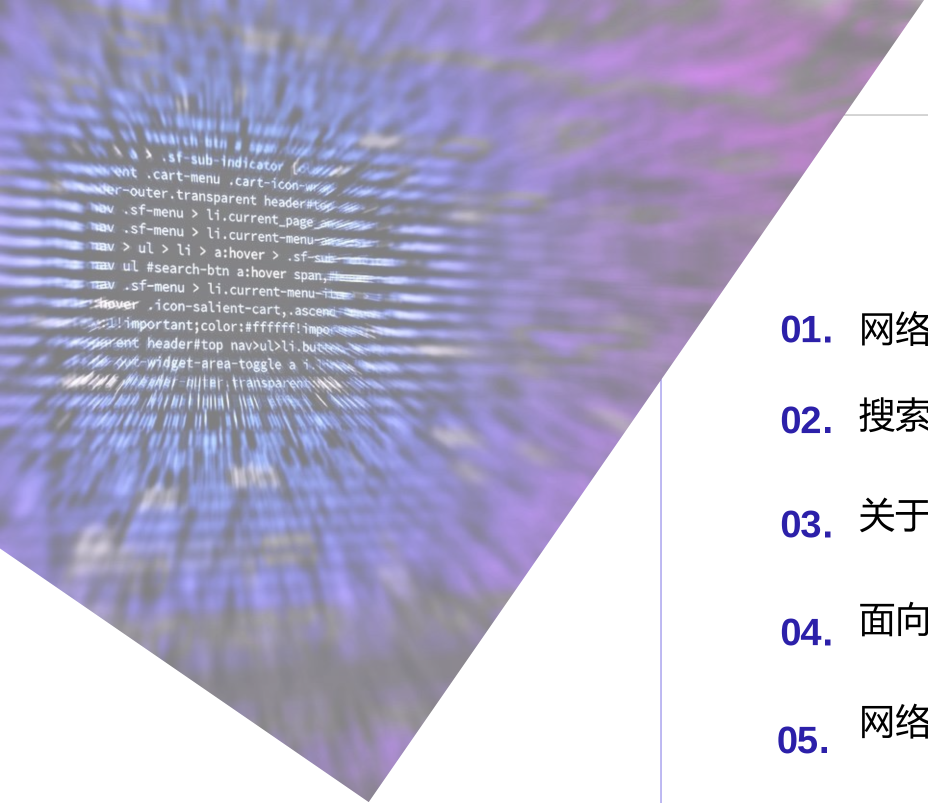
机器学习导论

第9章 网络机器学习

谢茂强

南开大学软件学院



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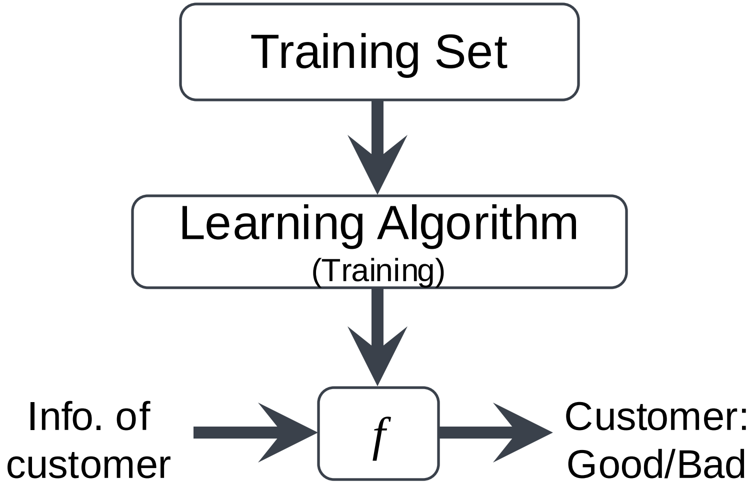
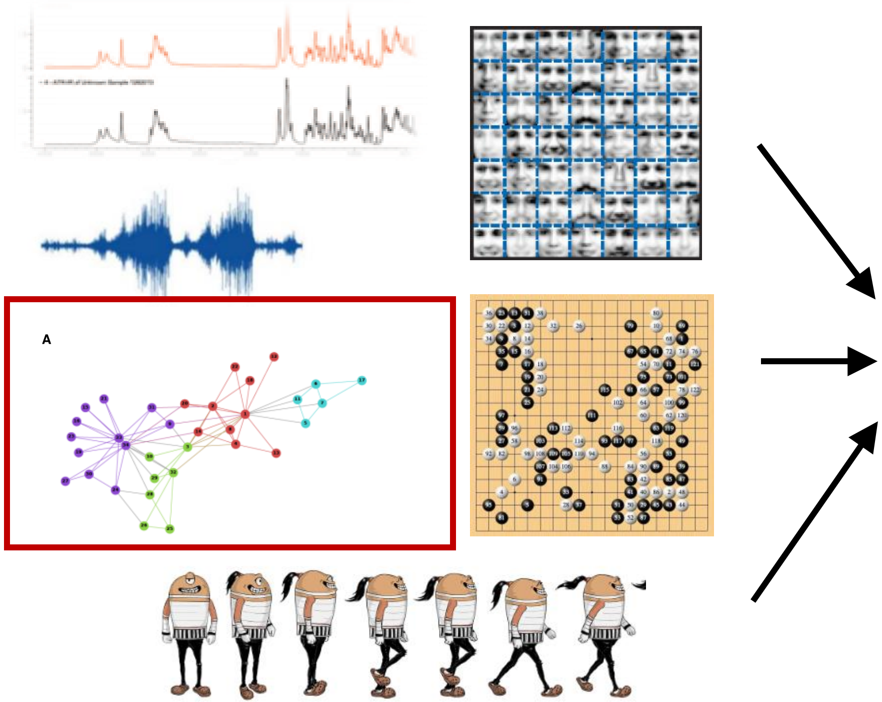
机器学习的范式

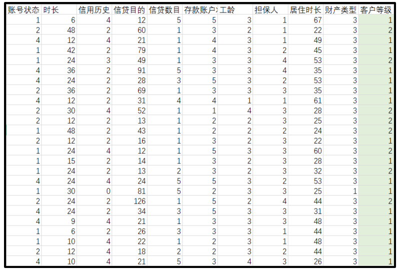


账号状态	时长	信用历史	信贷目的	信贷数目	存款账户	工龄	担保人	居住时长	财产类型	客户等级
1	6	4	12	5	5	3	1	67	3	1
2	48	2	60	1	3	2	1	22	3	2
4	12	4	21	1	4	3	1	49	3	1
1	42	2	79	1	4	3	2	45	3	1
1	24	3	49	1	3	3	4	53	3	2
4	36	2	91	5	3	3	4	35	3	1
4	24	2	28	3	5	3	2	53	3	1
2	36	2	69	1	3	3	3	35	3	1
4	12	2	31	4	4	1	1	61	3	1
2	30	4	52	1	1	4	3	28	3	2
2	12	2	13	1	2	2	3	25	3	2
1	48	2	43	1	2	2	2	24	3	2
2	12	2	16	1	3	2	3	22	3	1
1	24	4	12	1	5	3	3	60	3	2
1	15	2	14	1	3	2	3	28	3	1
1	24	2	13	2	3	2	3	32	3	2
4	24	4	24	5	5	3	2	53	3	1
1	30	0	81	5	2	3	3	25	1	1
2	24	2	126	1	5	2	4	44	3	2
4	24	2	34	3	5	3	3	31	3	1
4	9	4	21	1	3	3	3	48	3	1
1	6	2	26	3	3	3	1	44	3	1
1	10	4	22	1	2	3	1	48	3	1
2	12	4	18	2	2	3	2	44	3	1
4	10	4	21	5	3	4	3	26	3	1

Dataset：德国某银行的信用卡用户评级数据
http://archive.ics.uci.edu/ml/datasets/Statlog+(German+Credit+Data)

机器学习的范式





账号状态	时长	信用历史	信贷目的	信贷数目	存款账户	工龄	担保人	居住时长	财产类型	客户等级
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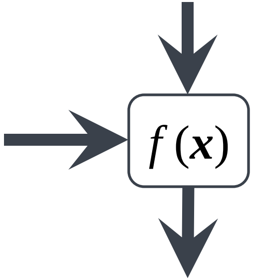
2	48	2	60	1	3	2	1	22	3	2
4	12	4	21	1	4	3	1	49	3	1
1	42	2	79	1	4	3	2	45	3	1
1	24	3	49	1	3	3	4	53	3	2
4	36	2	91	5	3	3	4	35	3	1
4	24	2	28	3	5	3	2	53	3	1
2	36	2	69	1	3	3	3	35	3	1
4	12	2	31	4	4	1	1	61	3	1
2	30	4	52	1	1	4	3	28	3	2
2	12	2	13	1	2	2	3	25	3	2
1	48	2	43	1	2	2	2	24	3	2
2	12	2	16	1	3	2	3	22	3	1
1	24	4	12	1	5	3	3	60	3	2
1	15	2	14	1	3	2	3	28	3	1
1	24	2	13	2	3	2	3	32	3	2
4	24	4	24	5	5	3	2	53	3	1
1	30	0	81	5	2	3	3	25	1	1
2	24	2	126	1	5	2	4	44	3	2
4	24	2	34	3	5	3	3	31	3	1
4	9	4	21	1	3	3	3	48	3	1
1	6	2	26	3	3	3	1	44	3	1
1	10	4	22	1	2	3	1	48	3	1
2	12	4	18	2	2	3	2	44	3	1
4	10	4	21	5	3	4	3	26	3	1

Training Dataset

x :Info. of customer



Customer: Good/Bad

SCI影响因子(IF)：关于科学论文的排序

Science，149(3683) : 510-515, July 30, 1965

Networks of Scientific Papers

The pattern of bibliographic references indicates the nature of the scientific research front.

Derek J. de Solla Price

This article is an attempt to describe in the broadest outline the nature of the total world network of scientific papers. We shall try to picture the network which is obtained by linking each published paper to the other papers directly associated with it. To do this, let us consider that special relationship which is given by the citation of one paper by another in its footnotes or

chine-handled citation studies, of large and representative portions of literature, which are much more tractable for such analysis than any topical indexing known to me. It is from such studies, by Garfield (*1, 2*), Kessler (*3*), Tukey (*4*), Osgood (*5*), and others, that I have taken the source data of this study.

percent of the papers contain no references at all; this notwithstanding, 50 percent of the references come from the 85 percent of the papers that are of the "normal" research type and contain 25 or fewer references apiece. The distribution here is fairly flat; indeed about 5 percent of the papers fall in each of the categories of 3, 4, 5, 6, 7, 8, 9, and 10 references each. At the other end of the scale, there are review-type papers with many references each. About 25 percent of all references come from the 5 percent (of all papers) that contain 45 or more references each and average 75 to a paper, while 12 percent of the references come from the "fattest" category—the 1 percent (of all papers) that have 84 or more references each and average about 170 to a paper. It is interesting to note that the number of papers with n references falls off in this "fattest" category as $1/n^2$, up to many hundreds per paper.

These references, of course, cover the entire previous body of literature. We can calculate roughly that, since the body of world literature has been

References

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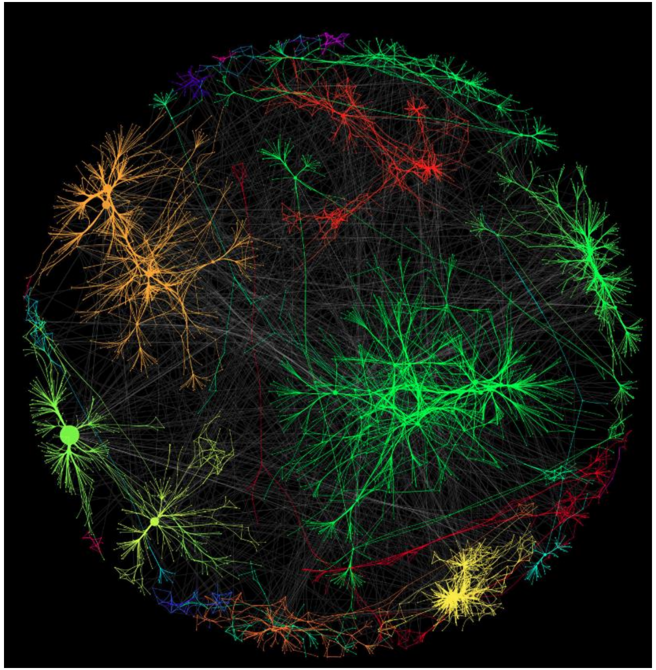
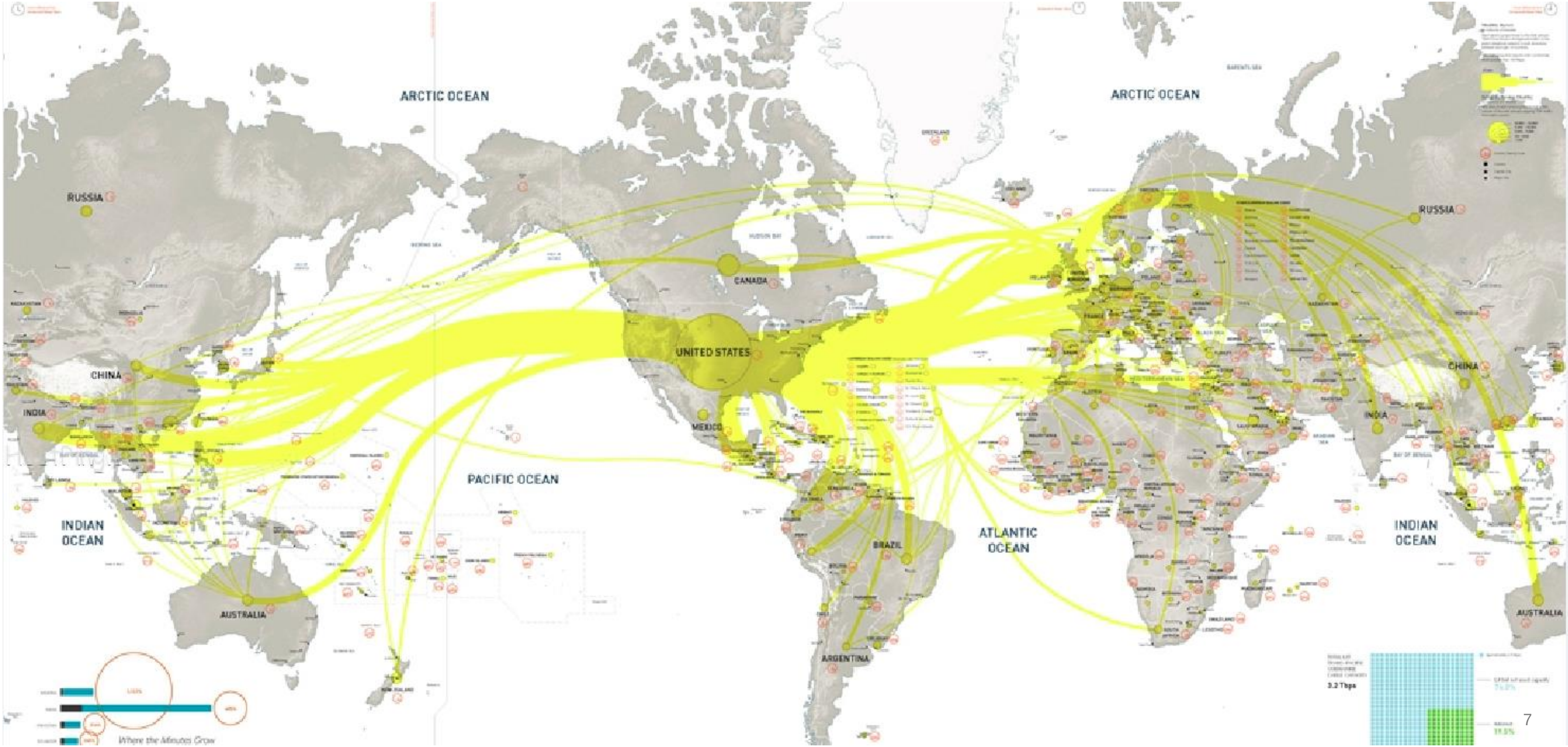
Prioritization / Ranking for Network

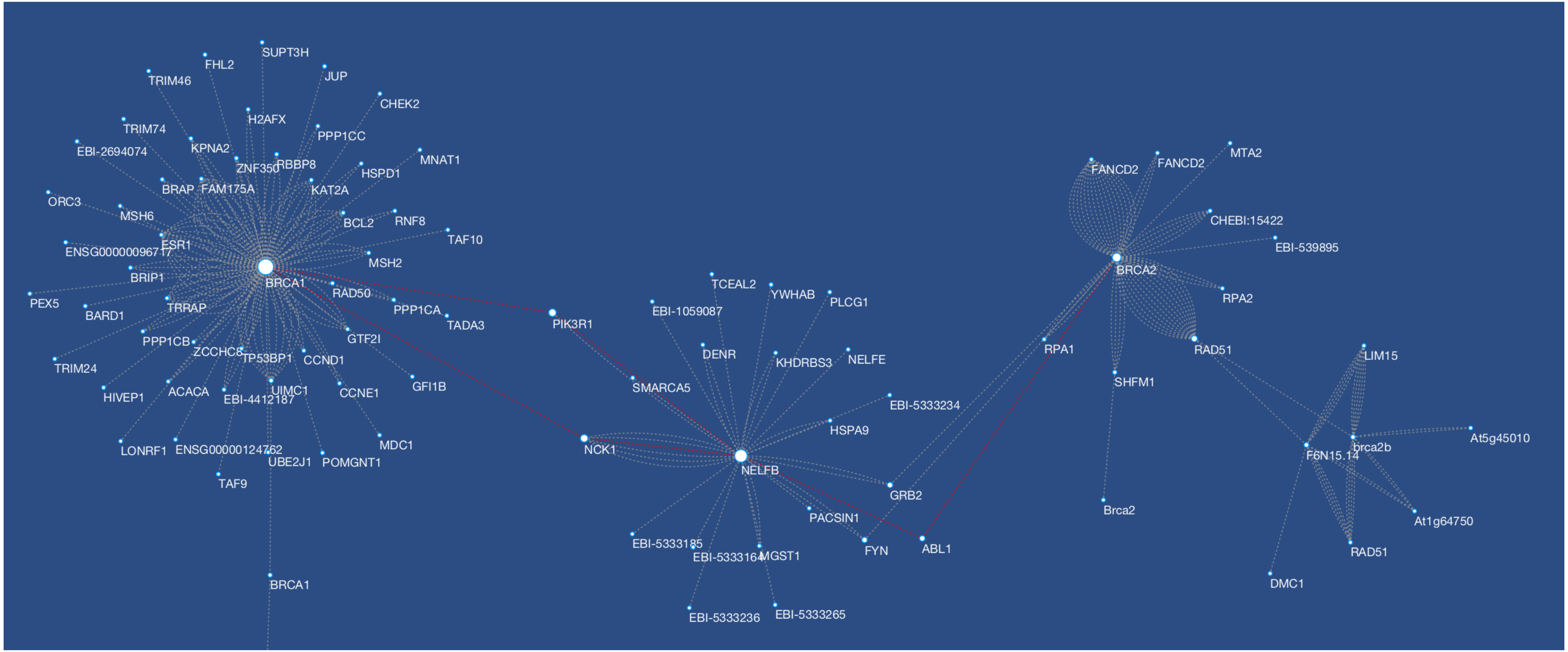
- 描述对象的特征 v.s 对象间的关系
  - SCI Impact Factors
  - PageRank for web retrieval
  - Information hub of social network
  - Disease causal gene or proteins



Internet & Web



基因网络


SUPT3H
FHL2
JUP
TRIM46
CHEK2
H2AFX
TRIM74
PPP1CC
EBI-2694074
KPNA2
ZNF350
RBBP8
MNAT1
HSPD1
BRAP
FAM175A
KAT2A
ORC3
MSH6
BCL2
RNF8
TAF10
ESR1
ENSG00000096717
MSH2
BRIP1
BRCA1
RAD50
PEX5
BARD1
TRRAP
PPP1CA
TADA3
PIK3R1
PPP1CB
ZCCHC8
TP53BP1
GTF2I
CCND1
TRIM24
HIVEP1
ACACA
EBI-4412187
UIMC1
CCNE1
GFI1B
LONRF1
ENSG00000124762
UBE2J1
POMGNT1
MDC1
TAF9
BRCA1
TCEAL2
YWHAB
PLCG1
EBI-1059087
DENR
KHDRBS3
NELFE
SMARCA5
EBI-5333234
HSPA9
NCK1
NELFB
GRB2
PACSIN1
EBI-5333185
EBI-5333164
MGST1
FYN
ABL1
EBI-5333236
EBI-5333265
FANCD2
FANCD2
MTA2
CHEBI:15422
EBI-539895
BRCA2
RPA2
RPA1
RAD51
SHFM1
LIM15
Brca2
At5g45010
brca2b
F6N15.14
At1g64750
RAD51
DMC1

例:使用搜索引擎搜索“南开大学”

1. 构建网站/网页间的关联网络
2. 类似于SCI IF，提出基于PageRank的网络排序算法

南开大学
https://www.nankai.edu.cn/main.htm
09月. 南开大学举行纪念"九一八"事变91周年系列活动. 17. 09月. 陈雨露校长调研马克思主义学院、化学学院. 17. 09月. 南开大学教授受邀在国际电化学会第73届学术年会作一小时大会报告. 17.

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南开新闻. 南开新闻网; 南开大学报; 院系机构. 专业学院; 职能部门; 直属和后勤单 ...

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研究生教育
预告：2022年南开大学-格拉斯哥大学双博士项... 关于2023年南开大学-格拉斯哥大 ...

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https://baike.baidu.com/item/南开
南开大学（Nankai University），简称"南开"，位于天津市，是中华人民共和国教育部直属的全国重点大学，是国家"双一流"建设高校，国家"211工程"和"985工程"建设高校，入选国家"珠峰计划" ...

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2 天前 · 1 教务处组织开展2022-2023学年秋季学期授课首日教学秩序检查. 2022-09-15. 2 "四新理念"框架下的新文科人才培养模式与机制研讨会举行. 2022-09-05. 3 南开大学面向2022级本科 ...

如何评价南开大学？ - 知乎
https://www.zhihu.com/question/318637180
2020-7-9 · 南开大学 研究生就业 国立西南联合大学 大学排名 世界一流大学和一流学科建设高校（双一流） 如何评价南开大学？为什么大家都在说南开没落了？但是我觉得南开在规模不占优 ...

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2022-9-19 · 南开大学 硕士研究生的基本修业年限一般为2至3年。硕士研究生的基本修业年限以其入学当年培养方案规定的学制为准。学术学位硕士研究生最长修业年限（含保留学籍和休学时 ...

南开大学举行纪念“九一八”事变91周年系列活动-南开要闻-南开 ...
news.nankai.edu.cn/ywsd/system/2022/09/18/030052832.shtml
2 天前 · 南开新闻网讯（通讯员阿依古丽张宏思记者乔仁铭摄影宗琪琪田宇洋吕双杨）9月18日，勿忘国耻复兴中华纪念九一八事变91周年系列活动在南开大学举行，南开师生齐聚国旗下、校钟 ...

【非凡十年 南开发展】润心铸魂 打造新时代“硬核”思政课 ...
news.nankai.edu.cn/ywsd/system/2022/09/19/030052851.shtml
1 天前 · 十年来，南开大学发挥马克思主义理论学科优势，聚力思想政治理论课建设，以特色课程为基石，培养时代新人，推动学校思想政治理论课改革创新不断迈上新台阶。……2012年以来

例: 网页搜索引擎（搜索+排序）

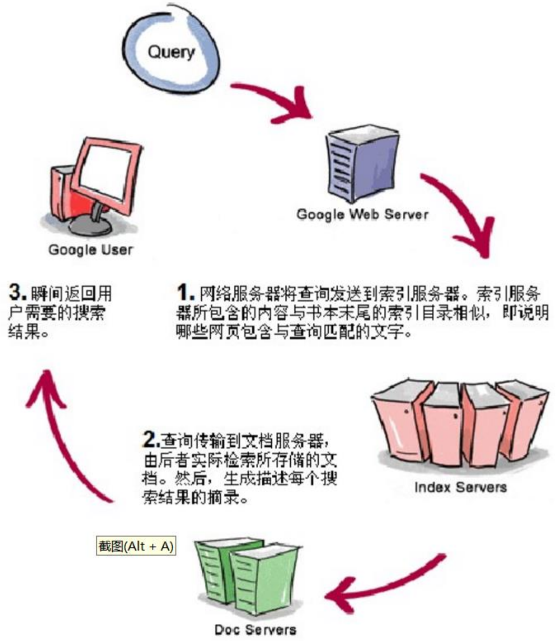
- 搜索“南开大学”
  - 搜索
    - 分词 “南开”、 “大学”
    - 根据建立的倒排索引，将同时包含“南开”和“大学”的文档返回
  - 排序
    - 根据内容相关性进行排序
    - 根据网络连接的权威程度

Google and PageRank

Sergey Brin & Lawrence Page, The PageRank Citation Ranking: Bring Order to the Web, 1998

利用网络拓扑结构对网页进行排序

类似的技术：SCI影响因子

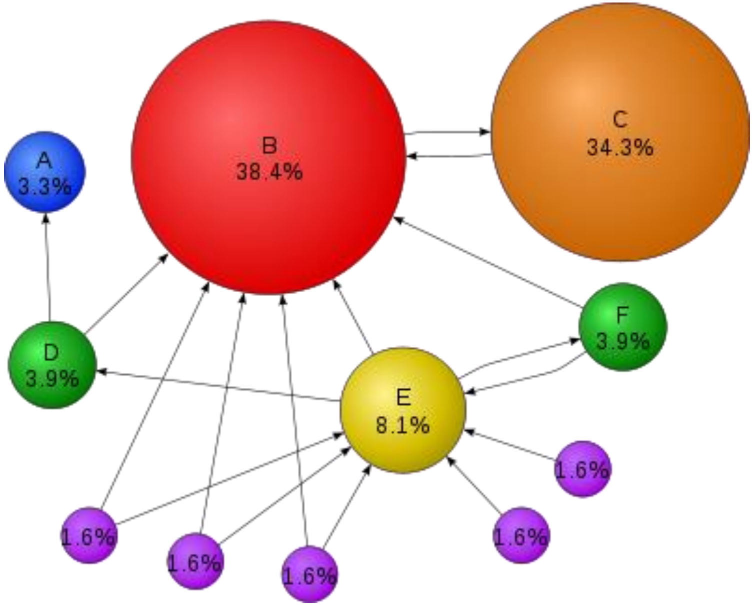


PageRank核心思想

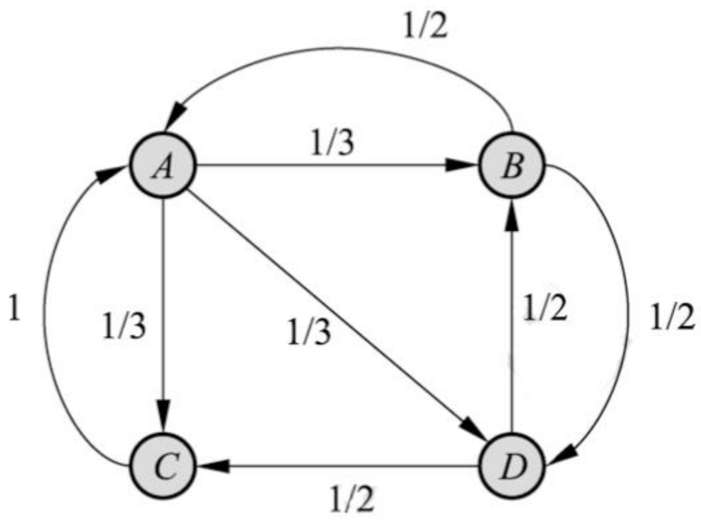
- Google把从A页面到B页面的链接解释为A页面给B页面投票;
  - 被连接多的页面权威度高;
  - 被权威度高的页面连接也会导致页面权威度高;
- 相对关系、递归定义

PageRank算法的难点

- 如何将相对权威度变成全局权威度
  - 通过随机游走模拟互联网用户群打开页面点击链接跳转页面（Random Surfing）
  - 通过随机游走，使页面跳转在全局达到稳定(平滑)
  - 认为游走者按照链接权重（概率）进行随机游走



随机游走

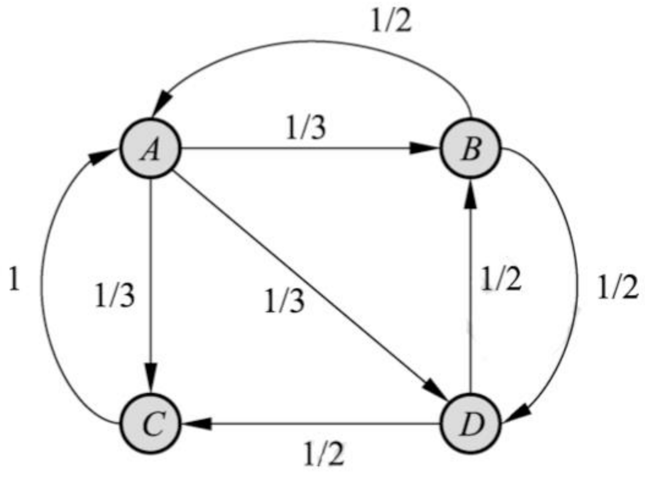


- 假设每个节点投票量一定,按链出等概率投出去

$$PR(A) = \frac{PR(B)}{2} + \frac{PR(C)}{1}$$

$$PR(A) = \frac{PR(B)}{L(B)} + \frac{PR(C)}{L(C)}$$

邻接矩阵(Affinity matrix)&转移概率矩阵



Source	Destination
A	B, C, D
B	A, D
C	A
D	B, C

邻接矩阵
(Affinity Matrix)

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

转移概率矩阵
(Transition Matrix)

$$\begin{bmatrix} 0 & 1/2 & 1 & 0 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{bmatrix}$$

基于随机游走的PageRank实现

计算所有页面p的$L(p)$，（由其倒数可构成转移概率矩阵）

Repeat

for all p_i

$$PR(p_i) = d \sum_{p_j \in M(p_i)} \frac{PR(p_j)}{L(p_j)} + (1-d)\frac{1}{N}$$

Until 收敛或达到最大迭代次数

N：节点个数，$M(p_i)$:链入p_i的页面集合，$L(p_j)$:p_j链出的页面

带重启的随机游走(Random Walk with Restart)

$$PR(p_i) = d \sum_{p_j \in M(p_i)} \frac{PR(p_j)}{L(p_j)} + (1-d)\frac{1}{N}$$

- 无外链页面会吞噬PageRank值
- d：继续访问下一个页面的概率
- 1-d：停止点击，随机浏览新页面

基于随机游走的PageRank实现(矩阵形式)

计算所有页面p的$L(p)$，（由其倒数可构成转移概率矩阵）

Repeat

$$\mathbf{R} = \begin{bmatrix} PR(p_1) \\ PR(p_2) \\ \vdots \\ PR(p_N) \end{bmatrix} = \begin{bmatrix} (1-d)/N \\ (1-d)/N \\ \vdots \\ (1-d)/N \end{bmatrix} + d \begin{bmatrix} \ell(p_1,p_1) & \ell(p_1,p_2) & \cdots & \ell(p_1,p_N) \\ \ell(p_2,p_1) & \ddots & & \\ \vdots & & \ell(p_i,p_j) & \\ \ell(p_N,p_1) & & & \ell(p_N,p_N) \end{bmatrix} \mathbf{R}$$

Until 收敛或达到最大迭代次数

$\ell(p_i,p_j)$ 从页面j指向页面i的链接数”与“页面j中含有的外部链接总数”的比值

其他的排序方法

- Text similarity between query and document
- Relevance between query and document
- Global ranking based on the topology of web
- Personal ranking by learning to rank

Learning with Local and Global Consistency

DengYong Zhou(周登勇), NIPS/NeurIPS 2003

- 提出可解释性很强的损失函数，兼顾经验风险（Global）和相邻元素的类别标签（Local）。
- 代价函数最优解 与RWR(Random Walk with Restart)算法迭代极限时的收敛值相同，换言之，为经典算法RWR给出了明确的代价函数，提升了RWR算法的理论价值。
- 亦可认为RWR作为所提损失函数的优化算法。

Learning with Local and Global Consistency中所提算法

1. Form the affinity matrix W defined by $W_{ij} = \exp(-\|x_i - x_j\|^2/2\sigma^2)$ if $i \neq j$ and $W_{ii} = 0$.
2. Construct the matrix $S = D^{-1/2} W D^{-1/2}$ in which D is a diagonal matrix with its (i, i)-element equal to the sum of the i-th row of W.
3. Iterate $F(t+1) = \alpha S F(t) + (1-\alpha) Y$ until convergence, where α is a parameter in $(0, 1)$.
4. Let F^* denote the limit of the sequence $\{F(t)\}$. Label each point x_i as a label $y_i = \arg\max_{j \leq c} F^*_{ij}$.

代价函数

3 Regularization Framework

Here we develop a regularization framework for the above iteration algorithm. The cost function associated with F is defined to be

$$\mathcal{Q}(F) = \frac{1}{2}\left(\sum_{i,j=1}^{n} W_{ij}\left\|\frac{1}{\sqrt{D_{ii}}}F_i - \frac{1}{\sqrt{D_{jj}}}F_j\right\|^2 + \mu\sum_{i=1}^{n}\left\|F_i - Y_i\right\|^2\right), \qquad (4)$$

Where $\mu > 0$ is the regularization parameter. Then the classifying function is

$$F^* = \arg\min_{F\in\mathcal{F}} \mathcal{Q}(F). \qquad (5)$$

使用相邻关系作为正则项

DengYong Zhou. Learning with Local and Global Consistency. NIPS 2013

代价函数的最优解

Differentiating $Q(F)$ with respect to F, we have

$$\left.\frac{\partial Q}{\partial F}\right|_{F=F^*} = F^* - SF^* + \mu(F^* - Y) = 0,$$

which can be transformed into

$$F^* - \frac{1}{1+\mu}SF^* - \frac{\mu}{1+\mu}Y = 0.$$

Let us introduce two new variables,

$$\alpha = \frac{1}{1+\mu}, \text{ and } \beta = \frac{\mu}{1+\mu}.$$

Note that $\alpha + \beta = 1$. Then

$$(I - \alpha S)F^* = \beta Y,$$

Since $I - \alpha S$ is invertible, we have

$$F^* = \beta(I - \alpha S)^{-1}Y.$$

带重启的随机游走迭代无穷多次后的收敛解

RWR算法迭代t次后的结果

$$F(t+1) = \alpha S F(t) + (1-\alpha) Y$$

$$F(t) = (\alpha S)^{t-1} Y + (1-\alpha) \sum_{i=0}^{t-1} (\alpha S)^i Y.$$

迭代次数趋近无穷时：

$$\lim_{t\to\infty} (\alpha S)^{t-1} = 0, \text{ and } \lim_{t\to\infty} \sum_{i=0}^{t-1} (\alpha S)^i = (I - \alpha S)^{-1}.$$

算法迭代无穷多次的收敛解：

$$F^* = \lim_{t\to\infty} F(t) = (1-\alpha)(I - \alpha S)^{-1} Y,$$

随机游走算法扩展至分类

F (t)从排序变为分类值

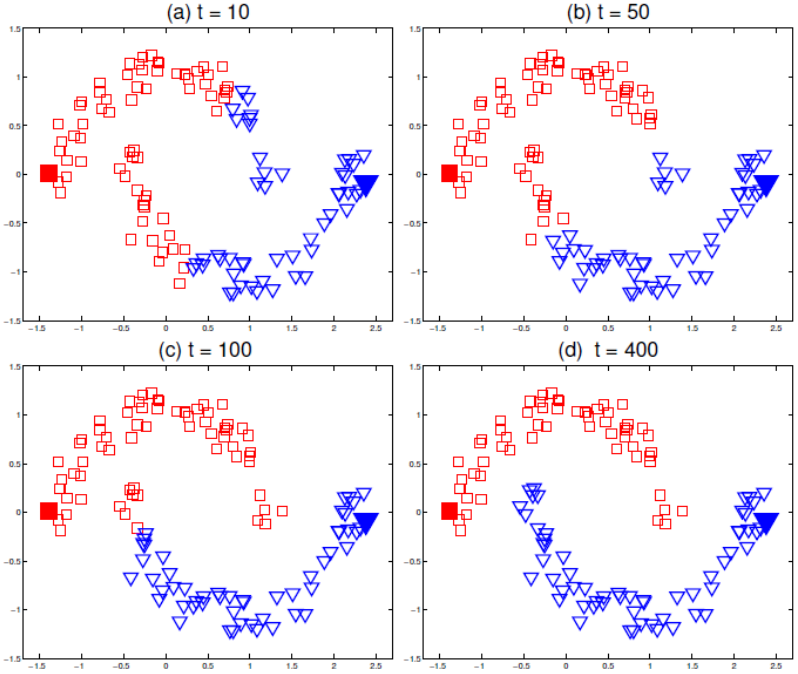


Figure 2: Classification on the pattern of two moons. The convergence process of our iteration algorithm with t increasing from 1 to 400 is shown from (a) to (d). Note that the initial label information are diffused along the moons.

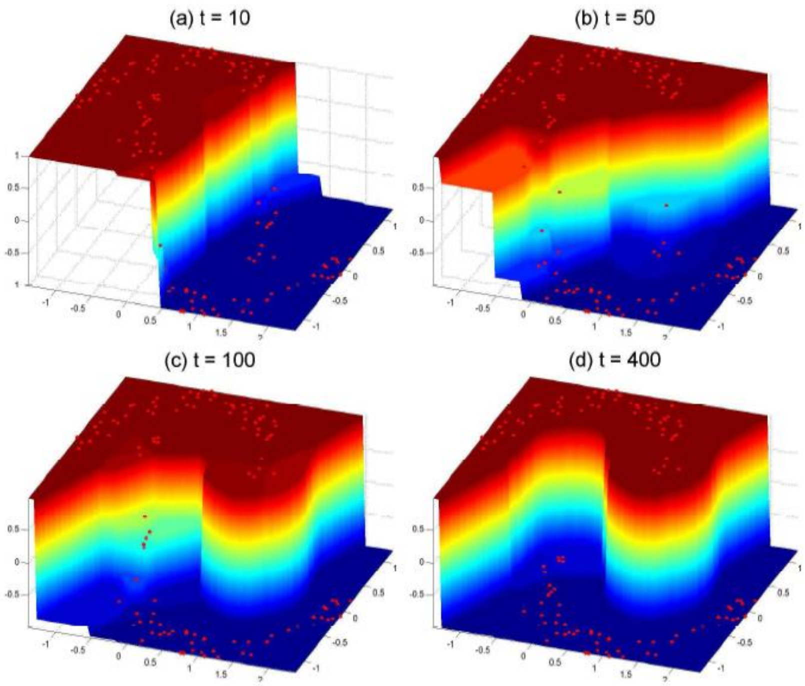
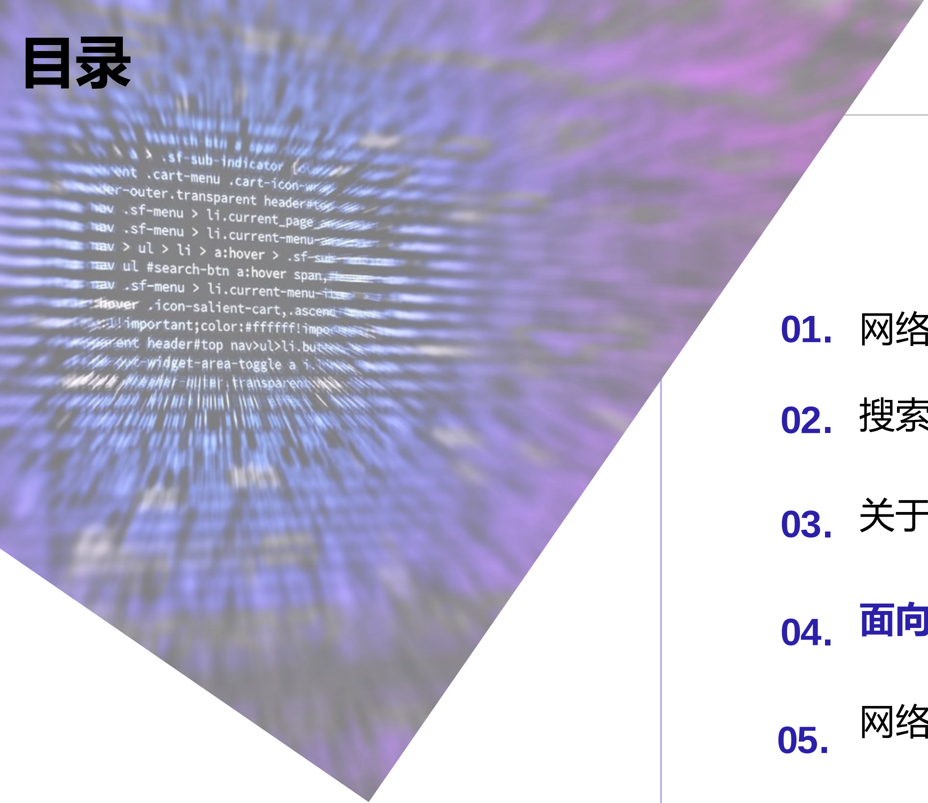


Figure 3: The real-valued classifying function becomes flatter and flatter with respect to the two moons pattern with increasing t. Note that two clear moons emerge in (d).



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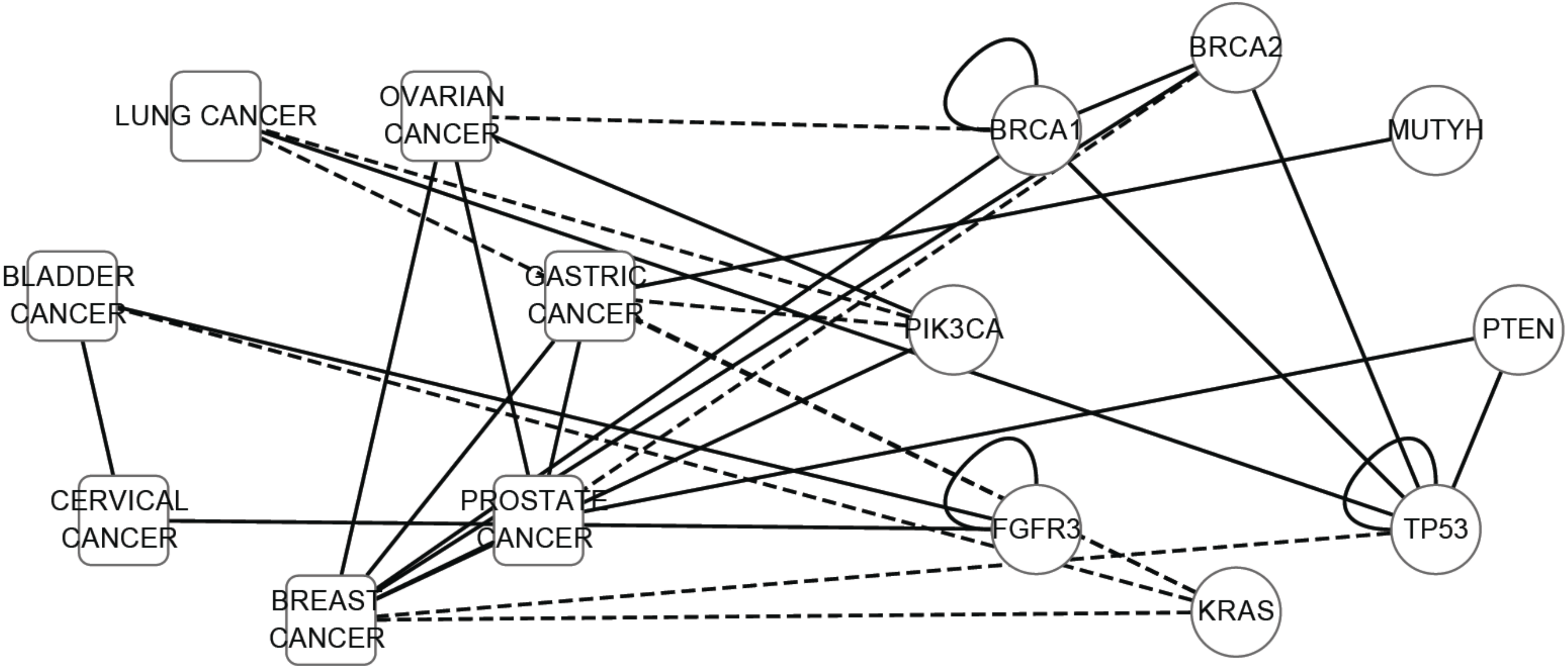
复杂网络(Phenotype-Gene Network)



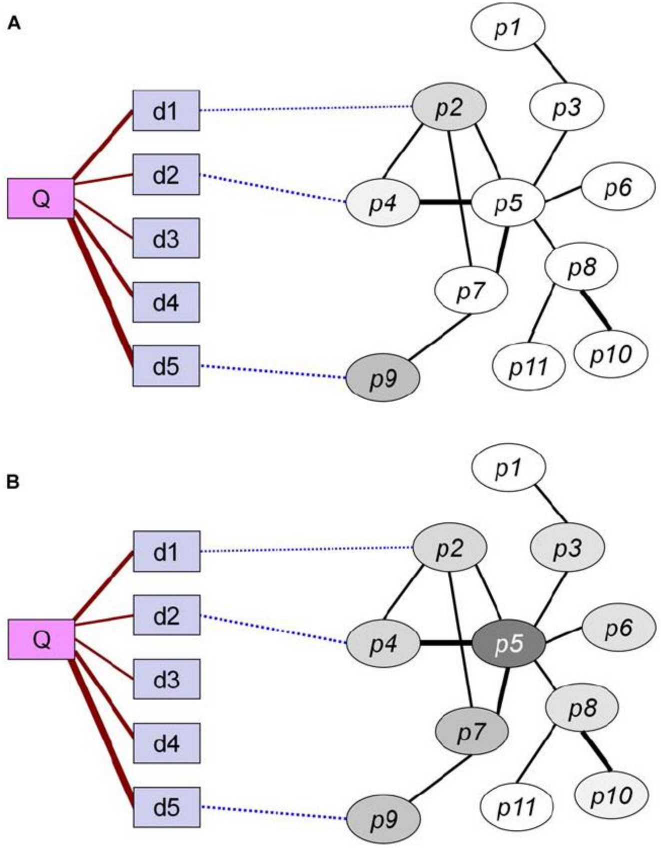
PRINCE

- 使用Random Walk with Restart进行排序受初始值很大影响（见DY Zhou的论文)
- 使用初始值作为Query，排序结果与Query相关
- 使用疾病的相似疾病的致病基因作为初始值，排序结果认为是与查询疾病最相关的致病基因候选集

O. Vanunu, Associating Genes and Protein Complexes with Disease via Network Propagation, PLoS Comp. Bio. 2010.

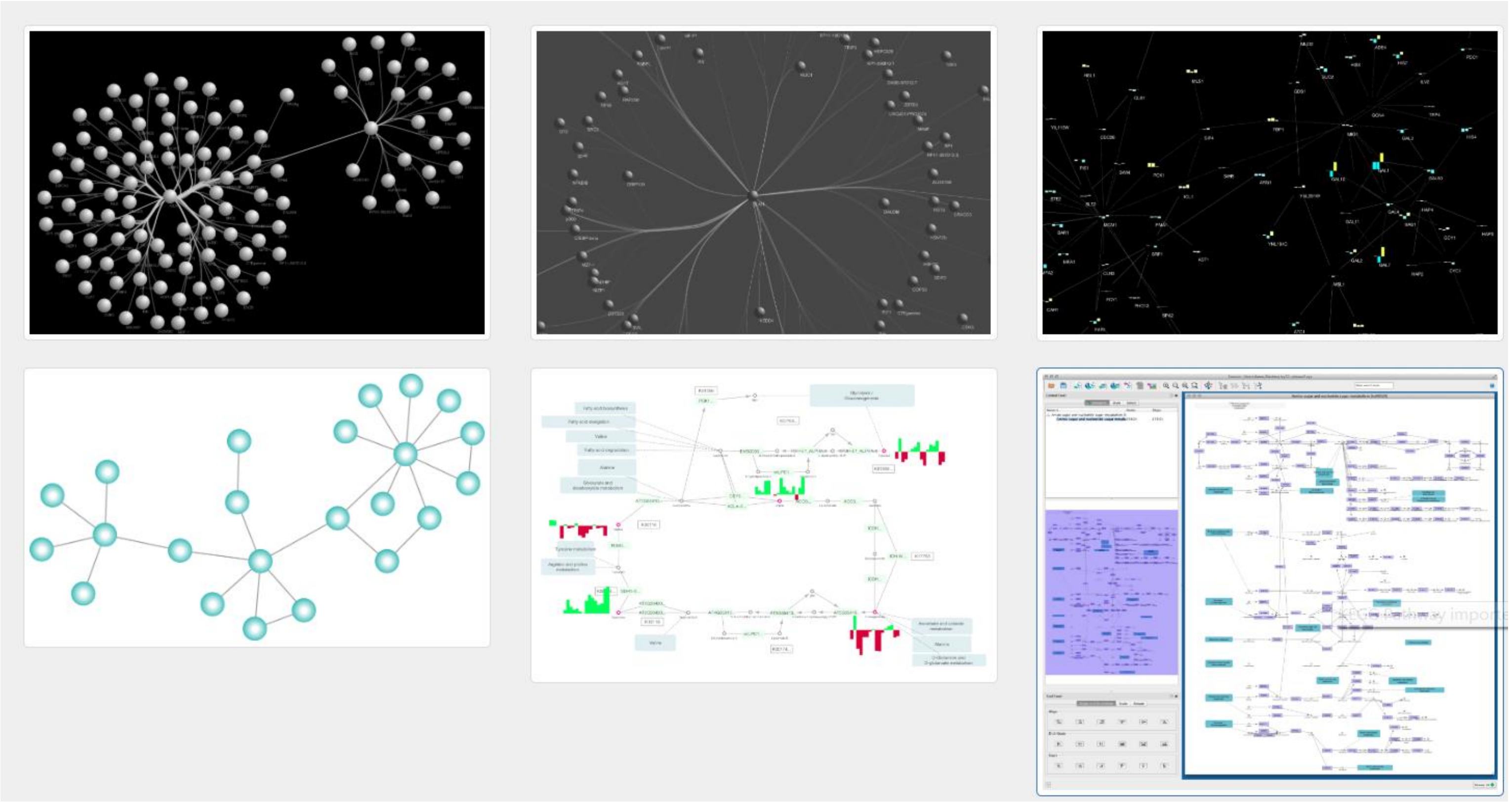
PRINCE for Prioritizing Genes

Figure 1. Illustration of the PRINCE algorithm. A query disease, denoted *Q*, has varying degrees of phenotypic similarity with other diseases, denoted *d1–d5* (marked with maroon lines, where thicker lines represent higher similarity). Known causal genes for these similar diseases are connected by dashed blue lines and used as the prior information. *p1–p11* comprise the protein set of a protein-protein interaction network, where interactions are marked with black lines and thicker lines denote edges with higher confidence. A scoring function that is smooth over the network is computed using an iterative network propagation method. At every iteration of the algorithm, each protein pumps flow to its neighbors and receives flow from them. Protein colors correspond to the flow they receive in a specific iteration, the darker the color the higher the flow. (A): the flow after the first iteration, representing the prior information. Only proteins *p2*, *p4* & *p9*, which are directly associated with similar diseases, have a positive incoming flow. (B): After several iterations, the amount of flow to each node converges, and the resulting flow, used to score the proteins, appears to be smooth over the network. *p5* emerges as the best causal gene candidate for disease *Q*, as it interacts with both *p2* and *p4*.



O. Vanunu, Associating Genes and Protein Complexes with Disease via Network Propagation, PLoS Comp. Bio. 2010.

使用CytoScape可视化网络



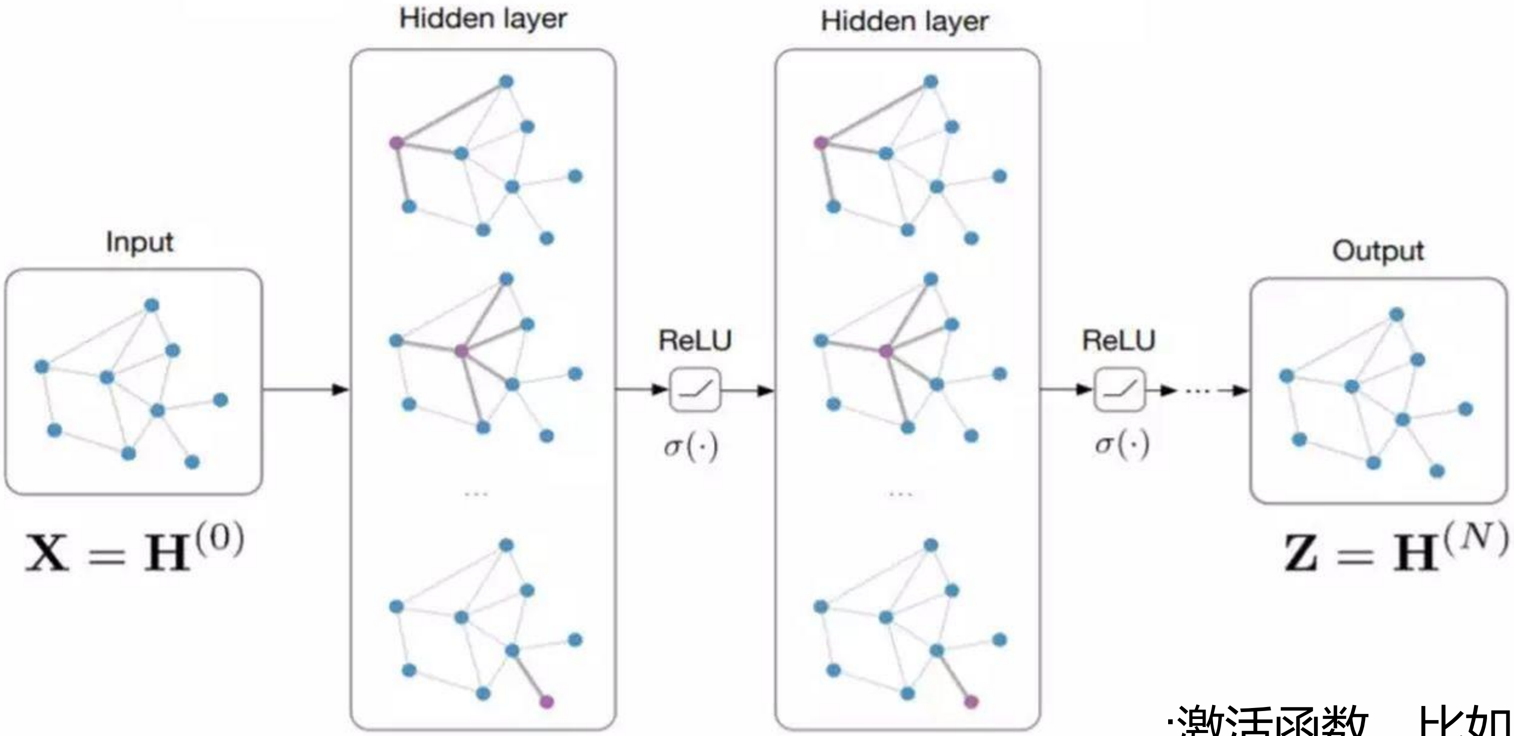
7. 网络表示学习或网络嵌入

Network Representation Learning, Network Embedding

- 深度神经网络本质上可以分成两部分
  - 特征构建部分（将节点间关联变换为节点特征）
  - 分类部分
- 关于特征构建
  - 图像、序列（文本）取得了重大成功
  - 在网络数据上（如社交网络）将节点给特征化可将深度学习推广到更广阔的的应用领域上。
  - 互联网推荐、生物信息网络
  - Graph Embedding、Network Embedding、Network Representation Learning

7. 使用K阶邻居表征当前节点

Input: Feature matrix $\mathbf{X} \in \mathbb{R}^{N \times E}$, preprocessed adjacency matrix $\hat{\mathbf{A}}$



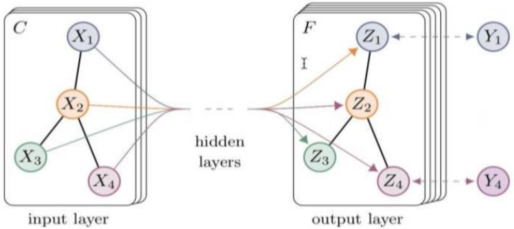
:激活函数，比如ReLU

$$\mathbf{H}^{(l+1)} = \sigma(\hat{\mathbf{D}}^{-\frac{1}{2}}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)})$$

7. 其他网络表示学习算法

GCN:

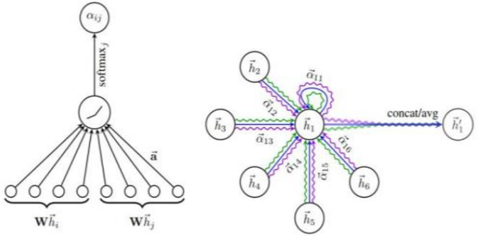
$$f\left(H^{(l)}, A\right) = \sigma\left(\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right)$$



GAT

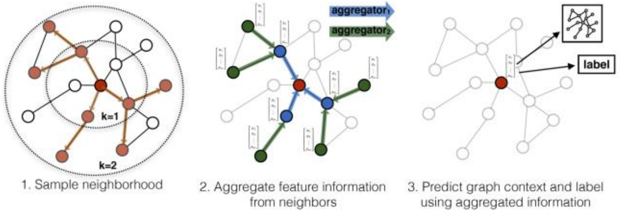
$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\vec{\mathbf{a}}^T[\mathbf{W}\vec{h}_i \| \mathbf{W}\vec{h}_j]\right)\right)}{\sum_{k\in\mathcal{N}_i} \exp\left(\text{LeakyReLU}\left(\vec{\mathbf{a}}^T[\mathbf{W}\vec{h}_i \| \mathbf{W}\vec{h}_k]\right)\right)}$$

$$\vec{h}'_i = \sigma\left(\sum_{j\in\mathcal{N}_i} \alpha_{ij}\mathbf{W}\vec{h}_j\right).$$



GraphSAGE

$\mathbf{h}_v^0 \leftarrow \mathbf{x}_v, \forall v \in \mathcal{V}$;
for $k = 1...K$ **do**
  for $v \in \mathcal{V}$ **do**
    $\mathbf{h}_{\mathcal{N}(v)}^k \leftarrow \text{AGGREGATE}_k(\{\mathbf{h}_u^{k-1}, \forall u \in \mathcal{N}(v)\})$;
    $\mathbf{h}_v^k \leftarrow \sigma\left(\mathbf{W}^k \cdot \text{CONCAT}(\mathbf{h}_v^{k-1}, \mathbf{h}_{\mathcal{N}(v)}^k)\right)$
  end
  $\mathbf{h}_v^k \leftarrow \mathbf{h}_v^k / \|\mathbf{h}_v^k\|_2, \forall v \in \mathcal{V}$
end
$\mathbf{z}_v \leftarrow \mathbf{h}_v^K, \forall v \in \mathcal{V}$



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